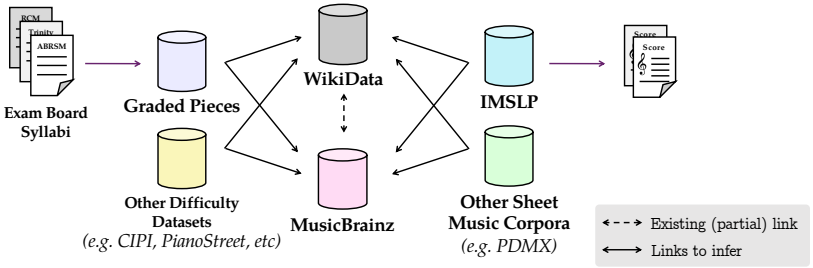
MetaMatch: Reconciling Metadata for Aggregating Music Difficulty Annotations

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Motivation

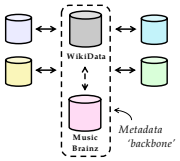
- Recent work in automated performance difficulty recognition focuses predominantly on piano music, due to data availability.
- Music exam board repertoire lists provide a rich source of difficulty labels for dozens of instruments; and the sheet music for many of the listed works is in the public domain.
- To make use of this data, we must first link musical works across multiple corpora with inconsistent metadata.

Application: Aggregating Music Difficulty Annotations



Method

- **Metadata backbone:** find a more complete matching by routing through *any* of several 'backbone' datasets, leveraging existing links between them for supervision.
- **Match composers:** first filter candidate works using *other* entities which are easier to reconcile (e.g. composers).
- **LLM-based entity recognition:** ask the LLM to classify (query, candidate) pairs as 'match', 'no match', or 'piece A is a part of piece B' (or vice versa).
- **Evaluation dataset:** to assess LLM accuracy, we label 447 graded pieces with their closest match in MusicBrainz.

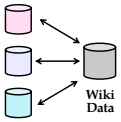


Dataset

- **58,104 repertoire list entries** collated from exam boards, covering **20+ instruments** and **13k+ composers**.
- Includes **7,737 entries** for piano.

Previous work

- The **LinkedMusic** project [1] uses OpenRefine's reconciliation API to semi-automatically match as many values to WikiData entities as possible.
- Peeters & Bizer [2] showed that LLMs with little or no fine-tuning perform competitively on entity matching tasks.



(Some) Challenges

Ambiguous composers

Query: "Jean-Baptiste Loeillet" or "Loeillet"

WikiData	
ID	**Label**
Q1319135	Jean-Baptiste Loeillet of London
Q1347001	Jean Baptiste Loeillet of Ghent

Composer names can be under-specified or ambiguous.

Alternative titles

Graded Pieces	
Title	Chaconne
Composer	Handel
Book	*Classics to Moderns*, Book 4
Difficulty	MTB, Grade 4, Piano

IMSLP	Match?
Suite in G minor, HWV 432	✓
Chaconne in F major, HWV 485	✗
Chaconne in G major, HWV 435	✗
Chaconne in G major, HWV 442	✗

Pieces can have different titles, including **translations and levels of detail**. For example, *Chaconne* here refers to movement *VI. Passacaglia* of HWV 432.

Different granularities

Graded Pieces	
Title	In Yugoslav Mode
Composer	Béla Bartók
Book	*MTB Piano Grade 1*
Difficulty	Grade 1

is part of → **IMSLP**: Mikrokosmos, Sz. 107

is the same piece as → **MusicBrainz**: Mikrokosmos, Sz. 107, BB 105: Volume II, No. 40: In Yugoslavian Mode (Allegretto)

No consistent meaning of 'musical work' across database schemas.

Missing entities

Query: "Tom Davoren"

WikiData
No entries

MusicBrainz.Artists	
ID	**Name**
d8acb7b0…	Tom Davoren
527f3f97-7…	Tony Davoren
57c6f649…	Tom Jones

Many musical works, and even artists, have no WikiData entry; **no single 'backbone' dataset covers all entities** one might wish to link.

Results

How many pieces can be matched?

Of 5,057 piano pieces with known composers:.

	MB.Works		IMSLP.Works	
No match	1,753	(34.7%)	2,668	(52.8%)
Any match	3,304	(65.3%)	2,389	(47.2%)
Exact	2,331	(46.1%)	954	(18.9%)
Part of	782	(15.5%)	1,427	(28.2%)
Contains	191	(3.8%)	8	(0.2%)

47% have a matching IMSLP work.

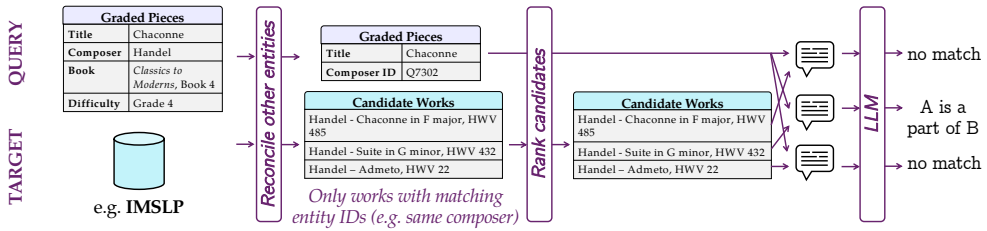
How many composers can be matched?

	GP.Composers	
Size	13,583	
Matched to WD	6,624	(48.8%)
Matched to MB	11,169	(82.2%)
Matched to both	6,116	
No MB→WD link	1,937	
Link agrees	3,615	(86.5%)
Link disagrees	564	(13.5%)

11k composers matched to a MusicBrainz artist.

Matching to MusicBrainz vs WikiData is **largely consistent.**

Matching Pipeline



Conclusion

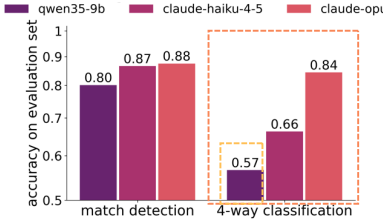
- Present method for reconciling entities across heterogenous music datasets, combining existing reconciliation services with LLM-based entity matching.
- Demonstrate that this method can be used to link entries from **music exam board repertoire lists to IMSLP works**, and hence **produce new difficulty-labelled datasets for multiple instruments.**

References

1. I. Fujinaga, "Sustainable archiving of music databases through RDF and NLQ2SPARQL," in *Proceedings of the 12th International Conference on Digital Libraries for Musicology*, 2025, pp. 100–104.
2. R. Peeters and C. Bizer, "Using ChatGPT for entity matching," in *European Conference on Advances in Databases and Information Systems*. Springer, 2023, pp. 221–230.

How accurate is the LLM matching?

LLMs can accurately detect when two pieces match (i.e. 'match' vs 'no match'), but **not the type of match** (i.e. 'match' vs 'A is part of B').



Commercial models substantially **outperform SOTA open weight models.**